* Lasso Regression = (L1 Regularization)

-> LASSO stands for Least Absolute Shrinkage and Selection operator.

- Primary goal is to find the balance model simplicity and accuracy.
- This model uses shrinkage. Shrinkage is where data values are shrunk towards a central point as the mean.
- The lasso procedure encourage simple, sparse models (model with fewer parameters).
- LASSO Regression is when we add penalty term to standard Linear regression based on the absolute values of the coefficients.
- The L1 Regularization term is the sum of the absolute values of the coefficients multipled by a tunning parameter λ.

$$L_1 = \lambda(B_1 + B_2 + B_3 + \dots B_n)$$

Linear Regression = $y = B_1x_1 + B_2x_2 + B_nx_n + c$

$$\text{Lasso} = (h_\theta(x) - y^\wedge)^2 + \lambda|\text{slope}|$$

$$\therefore \text{slop} = B_1 + B_2 + B_n$$

if $B_{1..n}$ is very small then respective features are neglated during model training. hence it helps in feature ~~st~~ selection.

- variables with 0 or small coefficients are removed from the model.

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